

Non-Functional Requirements Specification (NFRS)

Project: MoFlow: Smart Recurring Bill Payments Platform

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1. Introduction

1.1 Purpose

This document defines the non-functional requirements (NFRs) for the MoFlow Smart Recurring Bill Payments Platform. These requirements specify the quality attributes, operational characteristics, and constraints that ensure the platform is secure, reliable, performant, and scalable.

Unlike functional requirements, which describe what the system must do, non-functional requirements describe how well the system must perform.

1.2 Scope

The requirements in this document apply to all components of the MoFlow platform, including:

- Mobile Application
- Backend Services
- Payment Processing Engine
- Notification Service
- Administrative Portal
- Database
- APIs
- Cloud Infrastructure

2. Non-Functional Requirements

2.1 Performance Requirements

NFR-001 Response Time

The system shall respond to user requests within 2 seconds under normal operating conditions.

Priority: High

NFR-002 Payment Processing

Scheduled payment processing shall begin within 30 seconds of the scheduled execution time.

Priority: High

NFR-003 Dashboard Loading

The dashboard shall load within 3 seconds.

Priority: High

NFR-004 Search Performance

Payment history searches shall return results within 2 seconds.

Priority: Medium

NFR-005 Notification Delivery

Payment notifications shall be delivered within 60 seconds after payment completion.

Priority: High

2.2 Availability Requirements

NFR-006 System Availability

The platform shall maintain 99.9% monthly uptime, excluding scheduled maintenance.

Priority: High

NFR-007 Planned Maintenance

Scheduled maintenance shall occur during off-peak hours and be communicated to users in advance.

Priority: Medium

NFR-008 Fault Recovery

Critical services shall automatically recover after unexpected failures where technically feasible.

Priority: High

2.3 Reliability Requirements

NFR-009 Payment Reliability

Scheduled payments shall not be executed more than once for the same billing cycle.

Priority: High

NFR-010 Transaction Integrity

Every payment transaction shall be recorded accurately and completely.

Priority: High

NFR-011 Data Consistency

Payment records shall remain consistent across all system components.

Priority: High

2.4 Security Requirements

NFR-012 Data Encryption

Sensitive data shall be encrypted both in transit and at rest.

Priority: High

NFR-013 Secure Authentication

All users shall authenticate using secure authentication mechanisms.

Priority: High

NFR-014 Session Timeout

Inactive user sessions shall automatically expire after 15 minutes of inactivity.

Priority: High

NFR-015 Access Control

Users shall access only the information and functions authorized for their role.

Priority: High

NFR-016 Audit Logging

All security-sensitive activities shall be logged for auditing purposes.

Priority: High

NFR-017 Password Security

Passwords shall comply with defined security policies and be stored using secure hashing algorithms.

Priority: High

2.5 Scalability Requirements**NFR-018 Concurrent Users**

The system shall support at least 10,000 concurrent users without significant performance degradation.

Priority: Medium

NFR-019 Transaction Growth

The platform shall support increasing transaction volumes without requiring major architectural changes.

Priority: Medium

2.6 Usability Requirements

NFR-020 Ease of Use

New users shall be able to schedule their first recurring payment with minimal guidance.

Priority: High

NFR-021 Consistent Interface

The application shall maintain a consistent user interface across all screens.

Priority: Medium

NFR-022 Mobile Responsiveness

The application shall provide an optimized experience across supported mobile devices.

Priority: High

NFR-023 Error Messages

System error messages shall clearly explain issues and recommend corrective actions.

Priority: High

2.7 Accessibility Requirements

NFR-024 Accessibility Standards

The application shall comply with recognized accessibility guidelines where applicable.

Priority: Medium

NFR-025 Readability

The interface shall use readable fonts, sufficient contrast, and accessible color combinations.

Priority: Medium

2.8 Compatibility Requirements

NFR-026 Mobile Platforms

The application shall support current versions of Android and iOS.

Priority: High

NFR-027 Browser Compatibility

Administrative interfaces shall support major modern web browsers.

Priority: Medium

2.9 Maintainability Requirements

NFR-028 Modular Architecture

The system shall be designed using modular components to simplify maintenance.

Priority: Medium

NFR-029 Documentation

Technical documentation shall be maintained throughout the project lifecycle.

Priority: Medium

NFR-030 Logging

System events and application errors shall be logged for troubleshooting purposes.

Priority: High

2.10 Monitoring Requirements

NFR-031 Performance Monitoring

System performance metrics shall be continuously monitored.

Priority: Medium

NFR-032 Failure Alerts

Critical failures shall generate alerts for system administrators.

Priority: High

2.11 Backup and Recovery

NFR-033 Automated Backup

Application data shall be backed up automatically at scheduled intervals.

Priority: High

NFR-034 Disaster Recovery

The system shall support recovery following major failures or disasters.

Priority: High

2.12 Compliance Requirements

NFR-035 Regulatory Compliance

The platform shall comply with applicable financial and data protection regulations.

Priority: High

NFR-036 Data Privacy

User information shall be collected, processed, and stored in accordance with applicable privacy requirements.

Priority: High

2.13 Audit Requirements

NFR-037 Activity Tracking

Critical user actions shall be recorded in audit logs.

Priority: High

NFR-038 Administrative Actions

Administrative activities shall be fully traceable.

Priority: High

2.14 Interoperability Requirements

NFR-039 API Communication

The platform shall securely communicate with approved mobile money service provider APIs.

Priority: High

NFR-040 Third-Party Integration

External integrations shall use secure and documented interfaces.

Priority: Medium

3. Non-Functional Requirement Priorities

Priority	Description
High	Mandatory for MVP
Medium	Important for operational excellence
Low	Future enhancement

4. Verification Methods

Requirement Category	Verification Method
Performance	Performance Testing
Availability	Uptime Monitoring
Security	Security Testing & Penetration Testing
Reliability	Integration & System Testing
Scalability	Load Testing
Usability	Usability Testing
Accessibility	Accessibility Review
Backup & Recovery	Disaster Recovery Testing
Compliance	Compliance Audit

5. Assumptions

- Users have stable internet connectivity during payment scheduling.
- Mobile money providers expose reliable APIs.
- Cloud infrastructure supports required scalability.
- Users operate supported mobile devices.

6. Constraints

- Budget limitations may influence infrastructure choices.
- Third-party API availability affects payment processing.
- Regulatory requirements may evolve over time.
- Platform performance depends on external payment providers.

7. Conclusion

The Non-Functional Requirements Specification defines the quality standards that the MoFlow platform must achieve to deliver a secure, reliable, performant, and user-friendly recurring payment solution. These requirements complement the Functional Requirements Specification and provide measurable criteria for system architecture, development, testing, deployment, and operational support.